



V-OR

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FYP- Midterm Report

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October, 2025

Table of Contents

Introduction.....	5
1.1 Problem Statement.....	5
1.2 Scope.....	6
1.3 Modules.....	6
2.2.1 Module 1: Surgical Simulation Engine	6
2.2.2 Module 2: Virtual Training Mode.....	7
2.2.3 Module 3: Real-time Feedback and Performance Evaluation	7
2.2.4 Module 4: User Management System.....	7
1.4 User Classes and Characteristics	8
Project Requirements.....	9
2.1 Use-case Diagram.....	9
Extended Uses Cases:	10
Use Case 1 - Perform Surgery.....	10
Use Case 2 - Access Training Resources	11
Use Case 3 -Receive Guidance	12
Use Case 3 - Access 3D Anatomical Models	13
Use case 5 - Choose Surgery	15
2.2 Functional Requirements.....	16
USER MANAGEMENT SYSTEM:	16
SURGICAL SIMULATION ENGINE:	16
VIRTUAL TRAINING MODE:	16
REAL-TIME FEEDBACK & PERFORMANCE EVALUATION:	17
USER INTERFACE:	17
REPORTING AND ANALYTICS:	17
REPORTING AND ANALYTICS:	17
2.3 Non-Functional Requirements.....	17
SCALABILITY:	17
COMPLIANCE:	17
DATA INTEGRITY:	17
MAINTAINABILITY:	17
2.4 Domain Model.....	19
System Overview	20
3.1 Architectural Design	20
3.2 Design Models	20
• Class Diagram.....	21
• Class level Sequence Diagram.....	22
• Activity Diagrams.....	27

• State- Transition Diagram	32
3.3 Data Design	33
Databases and Data Storage:.....	33
ERD Diagram:	33
References.....	34

List of Figures

Figure 1 Source: Forum-Institut, <i>White Paper 01/2020: Sebastian</i> (2020)	6
Figure 2: Use case Diagram	10
Figure 3:Domain Model	20
Figure 4:Activity Diagram	22
Figure 5:Class Diagram	23
Figure 6: SD Perform Surgery	24
Figure 7: SD Access Training Resources	25
Figure 8: SD Receive Guidance	26
Figure 9: SD Access 3D Models	27
Figure 10: SD Surgery Selection	28
Figure 11: ERD Diagram	29

List of Tables

Table 1: User Classes and Characteristics	8
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Introduction

The purpose of this document is to outline the requirements for the Virtual Operating Room (V-OR), an advanced, highly interactive and immersive platform, in which users can sight and practice complex surgeries in a fully virtual environment. By utilizing the finest virtual reality, this V-OR platform has developed an extremely realistic and interactive medical domain in which the surgeries like pacemakers implant as well as many other sorts of operations can be performed with infinite repetitions. VOR is on a mission to change the way medical training is conducted by providing learners with access to a practice environment that not only can be practiced anytime and anywhere, but also without real danger. This document is intended for a diverse group of stakeholders, each of whom will benefit from the detailed requirements and specifications presented herein. The main target audiences are: Developers, Project Managers, Users, Testers k and Documentation Writers.

1.1 Problem Statement

In today's era it's hard for students in medicine to become skilled professionals due to lack of precision and accuracy in surgical training. Firstly, the biggest challenge they face is not having enough opportunities for hands-on practice, thus this leads to errors and inaccuracies during the real surgery. Additionally, the high cost of training tools such as physical simulators and other surgical training equipment makes it difficult for many schools, especially in areas with fewer resources, to afford them. Not only this, ethical concerns limit the use of cadavers (dead bodies), reducing opportunities for students to perform actual surgeries. As a result, students strive to build the confidence and skills necessary to perform the surgeries perfectly. This makes it hard for students to learn the procedures in depth as they are living in a controlled environment. It is important to address these issues so that future surgeons can perform procedures accurately and safely. The graph below illustrates the challenges that students face during medical training V-OR is a platform that aims to solve this by offering an affordable and ethical alternative for surgical training tools.

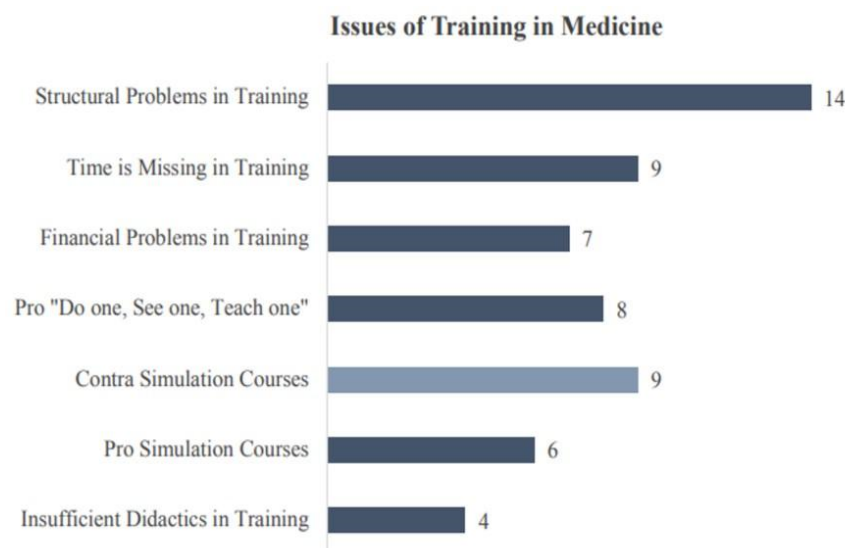


Figure 1
Source: Forum-Institut, White Paper 01/2020: Sebastian (2020)

1.2 Scope

Our V-OR aims to provide a highly interactive environment for surgeons to practice complex cardiac surgeries. By imitating real-life operating room conditions, it offers detailed 3D models, step-by-step guidance, and real-time feedback/suggestions to sweeten surgical skills without threat to patients.

The surgeries we will focus on include:

1. Atrial Septal Defect (ASD) Closure:

This procedure involves repairing a hole in the septum that separates the heart's two upper chambers (atria). Via our V-OR, surgeons will practice closing this genetic heart defect, which can cause blood to flow abnormally between the chambers, leading to difficulties like heart failure or stroke.

2. Pacemaker Implantation:

Pacemaker implantation is a life-saving strategy that concerns positioning a little electronic gadget under the skin to help restrain irregular heartbeats. Our V-OR will imitate the insertion of the pacemaker and its leads, linking them to the heart tissue. Surgeons will be trained to confirm the correct positioning of the pacemaker under the skin.

3. Coronary Artery Bypass Grafting (CABG):

CABG is one of the most typical and crucial surgeries for ministering coronary artery illness. This approach involves bypassing clogged or constricted arteries to fix normal blood flow to the heart. Our V-OR will provide instructions for graft harvesting (from the leg, arm, or chest), creating the bypass.

1.3 Modules

The V-OR platform consists of several modules, each designed to provide specific functionalities for medical training.

2.2.1 Module 1: Surgical Simulation Engine

This module involves creating a virtual environment for surgical training, which provides a realistic and interactive experience for users to fully utilize. This allows users to perform cardiac surgeries in a completely safe environment.

- **Feature 1:** Interactive 3D models of organs, tissues, and surgical tools.
- **Feature 2:** Step-by-step guidance to help users perform surgeries..
- **Feature 3:** Dynamic surgical scenarios to introduce real-world complexities.

2.2.2 Module 2: Virtual Training Mode

This module enables medical students and professionals to safely practice surgeries in a controlled environment. The users can either follow step-by-step guidance or practice without it. While practicing without guidance users will be evaluated based on how accurately they perform surgery on their own.

- **Feature 1:** Virtual training mode for practicing surgeries with or without step by-step guidance.
- **Feature 2:** Independent practice mode, where users can perform surgeries without any guidance and are evaluated on their accuracy and skills.

2.2.3 Module 3: Real-time Feedback and Performance Evaluation

This module provides immediate feedback to users while performing surgeries, helping them identify their mistakes in real-time. Furthermore, it also evaluates the user's performance throughout the surgery.

- **Feature 1:** Real-time error detection and then notifying users about the mistake.
- **Feature 2:** A performance evaluation system that will track the user's actions and gives a detailed feedback on their overall performance.

2.2.4 Module 4: User Management System

This module manages user profiles, tracks their progress, and gives access to training simulations. It also allows doctors or instructors to check student records and monitor their progress through detailed reports.

- **Feature 1:** User authentication and profile management.
- **Feature 2:** Tracks and saves each user's progress and performance.
- **Feature 3:** 3: Enables doctors or instructors to access student records and follow their training progress.

1.4 User Classes and Characteristics

Table 1: User Classes and Characteristics

User class	Description
Trainee	A medical student/first year surgeon performing high risk cardiac surgeries in the V-OR platform. Trainees physically engage through activities inside the virtual environment, with physical 3D anatomical models, surgical tools as well as guidance systems. They get feedback in terms of corrections on the performance they display and they make corrections on the errors. Trainees may perform surgeries that may include Atrial Septal Defect (ASD) closure, Pacemaker Implantation, or Coronary Artery Bypass Grafting (CABG). It is taken to monitor progress in order to aid the trainee in the improvement process. Estimated user base: 200-500 trainees.
Doctor	A supervisor of mid-level trainees, who is also an instructor and who assesses the performance of trainees and gives remarks. Doctors can see the training of trainees as well as their progress and the surgical outcomes that have been achieved. They can change scenarios of training so as to create new challenges for the students or help them to concentrate on certain skills. This platform is expected to be utilized by around fifty-one hundred doctors.
Simulation Admin	A technical staff member in charge of the management of the platform, software installation, and overseeing the virtual environments A 3D anatomical model is set up, and the operational capacity of the V-OR platform is Implementation of 3D anatomical models. Simulation Admins perform the maintenance of technical issues and explain to the users of the systems how to work with their systems.
3D Model Designer	A technical artist or developer focuses on developing and maintaining high definition and fidelity 3D anatomical models for the V-OR platform. They create accurate images of organs, tissues, and surgical instruments that will help in simulating real surgeries. Changes in the models may also be introduced by the designers, after receiving comments from the physicians or professionals in training
System Manager	A system administrator is a person who is closely related to the whole system in aspects of functions like user verification, data organization among others, and management of access over different users and resources. They assist in data management, tracking learning progress of the trainees and roles-based access to modules including the Surgical Simulation Engine, Virtual Training Mode as well as the Real Time Feedback System.

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case Diagram

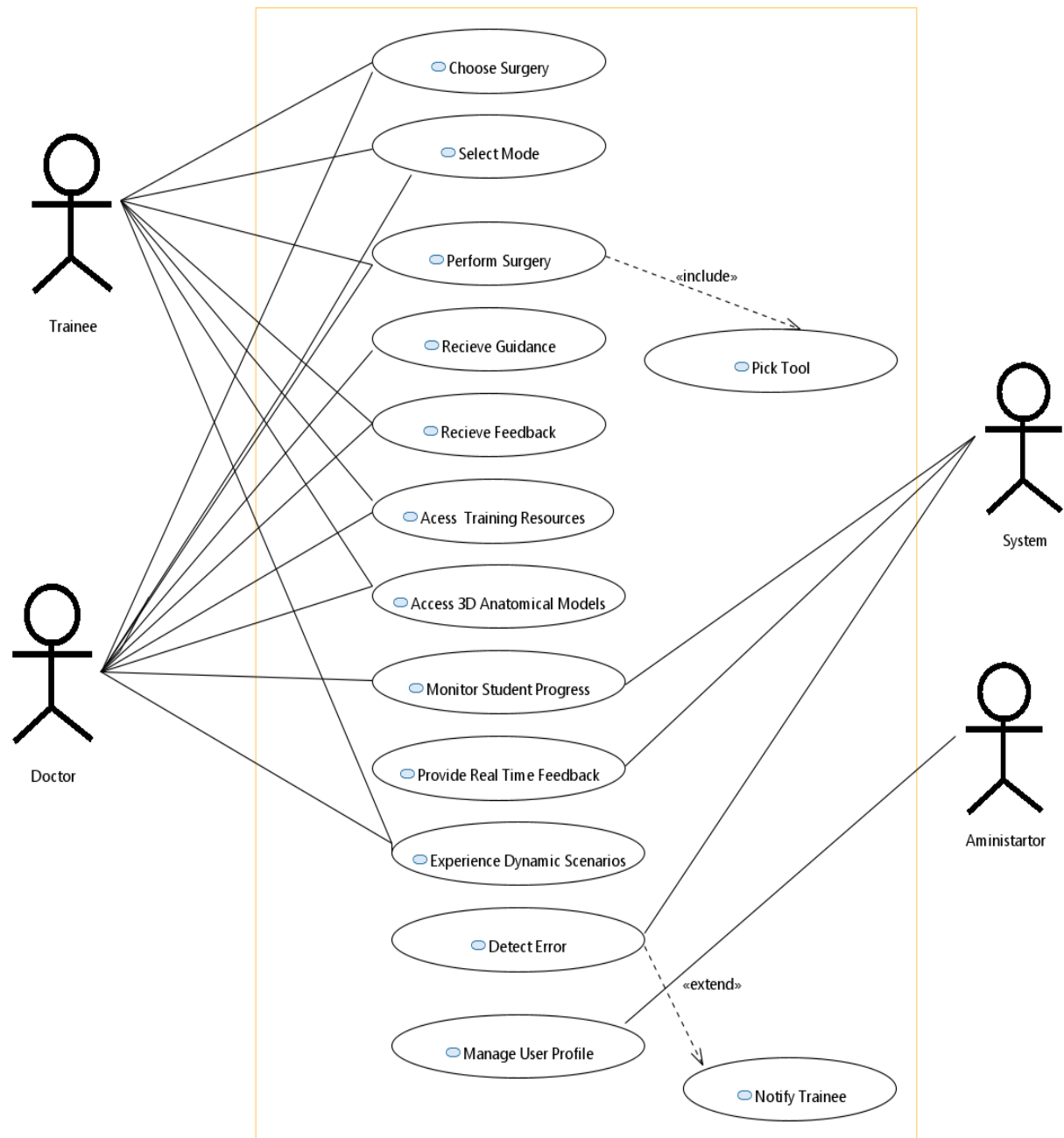


Figure 2: Use case Diagram

Extended Uses Cases:

Use Case 1 - Perform Surgery

Use Case	01	
Name	Perform Surgery	
Scope	V-OR Platform	
Level	User level goal	
Primary Actors	Trainee	
Stakeholders and interest	Trainee: Who wants to gain hands-on practice and improve skills. Doctor: Who want to further enhance their skills.	
Pre-Condition	• All required tools and resources are available • Trainee has selected mood and type of surgery	
Post-Condition	• The surgery has been completed • The trainee has gained feedback on it	
Main Success Scenario		
	User Action	System Response
	1.Trainee picks up the tools.	
		2.The system provide tools and 3D anatomical models.
	3.Trainee starts to perform surgery.	
		4.The system provides step by step guidance and real time error detection (if guidance is enabled).
	5.Trainee completes the surgery	
		6.The system saves progress and outputs performance.

Extensions	
Special Requirements	● VR headset to run the application ● High-fidelity 3D models. ● Feedback and Error Detection mechanism
Technology and Data Variation List	The VR headset needs to be Oculus or Quest.
Frequency of Occurrence	Frequently
Miscellaneous	N/A

Use Case 2 - Access Training Resources

Use Case	02	
Name	Access Training Resources	
Scope	V-OR Platform	
Level	User level goal	
Primary Actors	Trainee, Doctor	
Stakeholders and interest	Trainee: Who wants to learn about resources and equipment. Doctor: Who wants to guide trainees during training sessions.	
Pre-Condition	● Trainee has logged into the system. ● Training resources are available and updated.	
Post-Condition	The trainee has gained access to all the resources.	
Main Success Scenario		
	User Action	System Response
	1.Trainee selects option for training resources.	

		2.The system shows a list of all available resources.
	3.Trainee selects a specific and reviews it.	
		4.The system loads the specified resource.
Extensions		
Special Requirements	VR headset to run the application.	
Technology and Data Variation List	The VR headset needs to be Oculus or Quest.	
Frequency of Occurrence	Frequently	
Miscellaneous	N/A	

Use Case 3 -Receive Guidance

Use Case	03
Name	Receive Guidance
Scope	V-OR Platform
Level	User level goal
Primary Actors	Trainee
Stakeholders and interest	Trainee: Who wants to perform surgery with guidance. real time error detection and feedback. Doctor: Who wants to gain precision and improve skills.
Pre-Condition	Trainee has selected the guidance mood

Post-Condition	● Trainee completes the surgery with step by step guidance. ● The trainee was notified on error throughout.	
Main Success Scenario		
	User Action	System Response
	1.Trainee selects the guidance mood.	
		2.The system provides instructions and step by step guidance of the chosen surgery
	3.Trainee starts to perform surgery accordingly as guided.	
		4. The system provides in alerts for mistakes, and real-time error detection.
	5.Trainee completes the surgery successfully.	
Extensions	If the trainee makes the same wrong move all the time, the system can stop for the trainee to be taught the correct way to do it.	
Special Requirements	● VR headset to run the application ● High quality guidance (visual or speech) system to provide step by step instructions. ● Feedback and Error Detection mechanism	
Technology and Data Variation List	The VR headset needs to be Oculus or Quest.	
Frequency of Occurrence	Frequently	
Miscellaneous	N/A	

Use Case 3 - Access 3D Anatomical Models

Use Case	04
Name	Access 3D Anatomical Models

Scope	V-OR Platform	
Level	User level goal	
Primary Actors	Trainee, Doctor	
Stakeholders and interest	Trainee: Who learns about these models and performs surgeries on them. Doctor: Who want perform further experiments.	
Pre-Condition	3D anatomical models are available in the system with correct textures and sizes	
Post-Condition	• The surgery has been performed on that models • The trainee has viewed them carefully.	
Main Success Scenario		
	User Action	System Response
	1.Trainee views the anatomical models.	
		2.The system shows 3D anatomical models.
	3.Trainee interacts with the model before performing surgery.	
		4.The system allows interaction.
Extensions		
Special Requirements	VR headset to run the application High-fidelity 3D models. Feedback and Error Detection mechanism	
Technology and Data Variation List	The VR headset needs to be Oculus or Quest.	
Frequency of Occurrence	Frequently	

Miscellaneous	N/A
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Use case 5 - Choose Surgery

Use Case	05												
Name	Choose Surgery												
Scope	V-OR Platform												
Level	User level goal												
Primary Actors	Trainee												
Stakeholders and interest	Trainee: Who wants to select surgery that they want to perform.												
Pre-Condition	• The trainee has logged into the system. • The V-OR platform offers the list of available surgeries.												
Post-Condition	The trainee has selected surgery and mode.												
Main Success Scenario	<table> <tr> <th>User Action</th><th>System Response</th></tr> <tr> <td>1.Trainee select the “choose surgery” option.</td><td></td></tr> <tr> <td></td><td>2.The system shows a list of available surgeries (ASD Closure, Pacemaker Implantation, CABG).</td></tr> <tr> <td>3.Trainee selects a surgery from the list.</td><td></td></tr> <tr> <td></td><td>4.The system confirms the selection and asks for mode.</td></tr> <tr> <td>5.Trainee chooses the desired mode.</td><td></td></tr> </table>	User Action	System Response	1.Trainee select the “choose surgery” option.			2.The system shows a list of available surgeries (ASD Closure, Pacemaker Implantation, CABG).	3.Trainee selects a surgery from the list.			4.The system confirms the selection and asks for mode.	5.Trainee chooses the desired mode.	
User Action	System Response												
1.Trainee select the “choose surgery” option.													
	2.The system shows a list of available surgeries (ASD Closure, Pacemaker Implantation, CABG).												
3.Trainee selects a surgery from the list.													
	4.The system confirms the selection and asks for mode.												
5.Trainee chooses the desired mode.													

		6.The system sets up surgery environment.
Extensions		
Special Requirements	When selecting the types of surgery and modes, clear graphic of what is ongoing is established.	
Technology and Data Variation List	• The VR headset needs to be Oculus or Quest. • A desktop or VR application.	
Frequency of Occurrence	Frequently	
Miscellaneous	N/A	

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style.

USER MANAGEMENT SYSTEM:

- The system should allow users to login and manage their accounts.
- User roles should include students, surgeons and administrators.
- Users should be able to update their profiles, view their training process and feedback reports.
- The system should track and securely save the user's progress in different training modules.

SURGICAL SIMULATION ENGINE:

- The platform should include interactive 3D models of organs, tissues and surgical tools.
- The system should provide step by step guidance during the training surgeries.
- The dynamic scenarios generated by the system should be like real- world complexities.

VIRTUAL TRAINING MODE:

- Users should be able to take an assessment surgery where no help will be provided and at the end they will be evaluated on their performance.

- Users should be able to choose the surgery they want to perform from the 3 options we are currently providing; ASD, pacemaker and CABG.

REAL-TIME FEEDBACK & PERFORMANCE EVALUATION:

- The system should be able to detect real-time errors during the surgery and notify the user.
- The system should be able to provide detailed feedback and by using a predefined criterion, evaluate the user performance.

USER INTERFACE:

- The system should provide an intuitive dashboard for users to navigate to the selection of surgeries, type of surgeries and track their progress.

REPORTING AND ANALYTICS:

- The instructors should be able to review the feedback report of the students and their assessment reports.
- Instructors should be able to provide feedback to those students' reports.

REPORTING AND ANALYTICS:

- The system should be compatible with Oculus VR headset and hand sensors.

2.3 Non-Functional Requirements

This section specifies non-functional requirements.

SCALABILITY:

- The system will be scalable enough so that in future it can accommodate additional complex surgeries.

COMPLIANCE:

- The system will comply with medical training standards.
- The system should comply with data protection regulations such as GDPR for user data privacy.

DATA INTEGRITY:

- The system should ensure that all performance data is captured and stored without loss.

MAINTAINABILITY:

- The code should be modular and well documented, allowing future developers to easily add features.

- The system should include detailed documentation covering all major components and functionalities.

COMPATIBILITY:

- The VR environment should be compatible with multiple VR headsets, including Oculus Quest 2.
- The system should be compatible with major operating systems like Windows and Linux.

SECURITY:

- User authentication will be required for all access to the system, with role-based access control implemented for different user types (students, instructors, administrators).
- Sensitive data will only be accessible to authorised users.

RELIABILITY:

- The system will be highly reliable, with minimal downtime to ensure uninterrupted training.

AVAILABILITY:

- It will be available 24/7 to allow medical professionals to practise at any time.

USABILITY:

- The interface should be user-friendly and intuitive for medical students and professionals to navigate and use the virtual environment.
- The VR experience should be immersive and realistic, making the virtual surgeries as close to real-world scenarios as possible.

PERFORMANCE:

- The system should provide real-time feedback during surgeries without any noticeable delay.

2.4 Domain Model

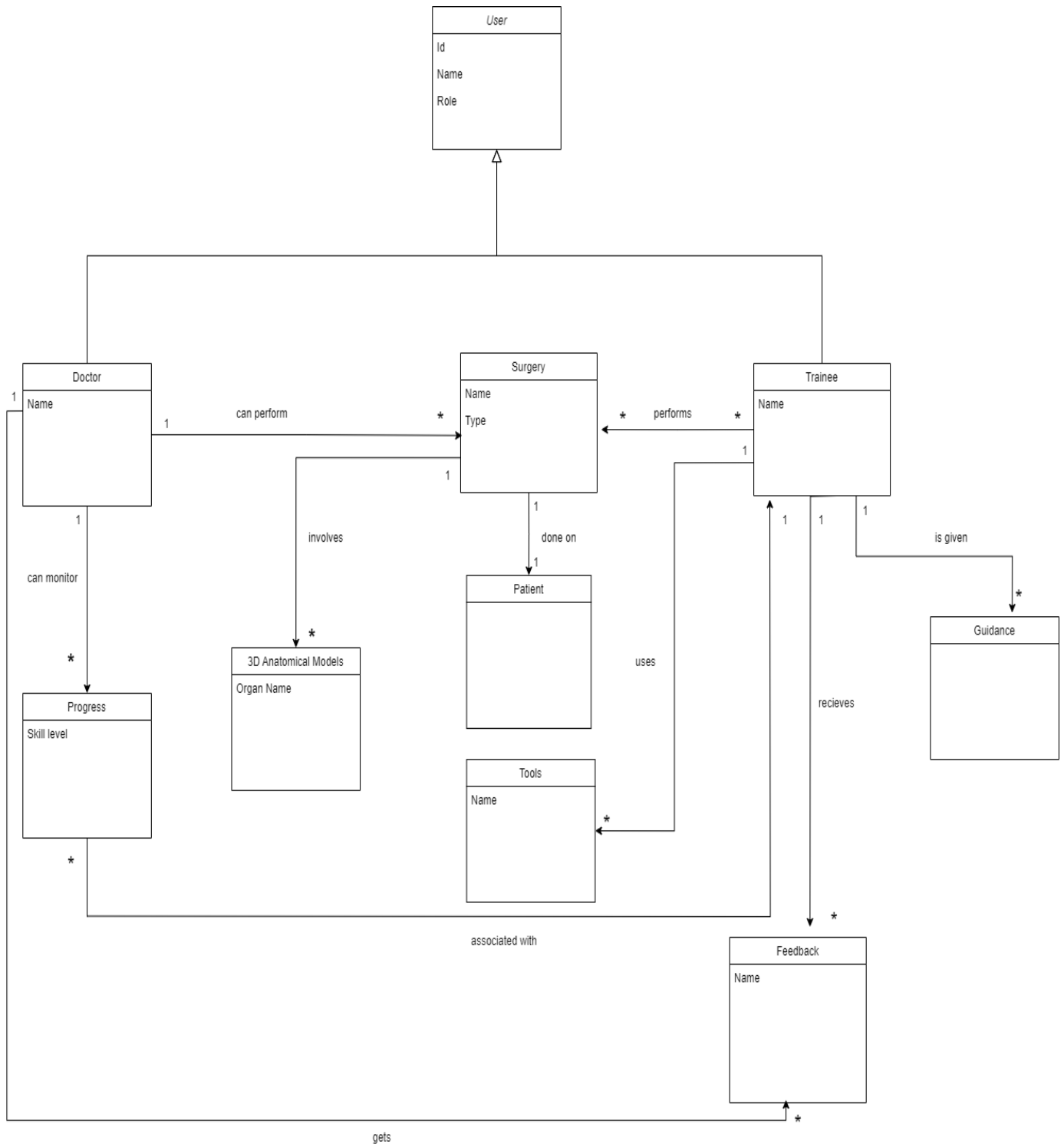
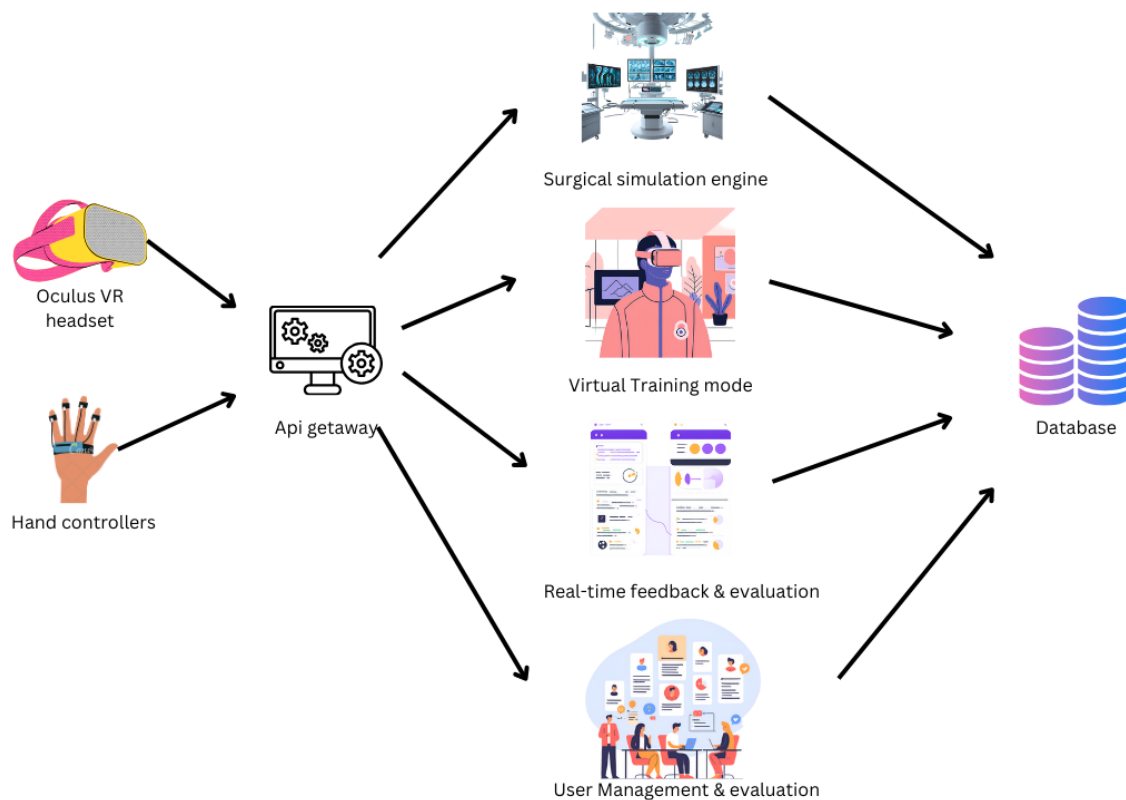


Figure 3: Domain Model

System Overview

3.1 Architectural Design

The Virtual Operating Room system lets medical trainees and surgeons practice surgeries in a VR environment using VR headsets and hand controllers. An API Gateway connects the user's device to the system's backend services. The Surgical Simulation Engine creates 3D models and offers step-by-step instructions. Users can choose to practice with or without these instructions, and the Real-Time Feedback system gives instant error detection and tracks performance. The User Management System takes care of logging in and tracking progress, while all information is stored in a central database. The project uses Git and GitHub to manage changes and help the team work together.



Architecture Diagram 1

3.2 Design Models

During the V-OR development, we are using the Object-Oriented Programming (OOP) approach. This methodology provides for the development of clear tangible classes and objects within the system in a more endearing approach.

- **Class Diagram**

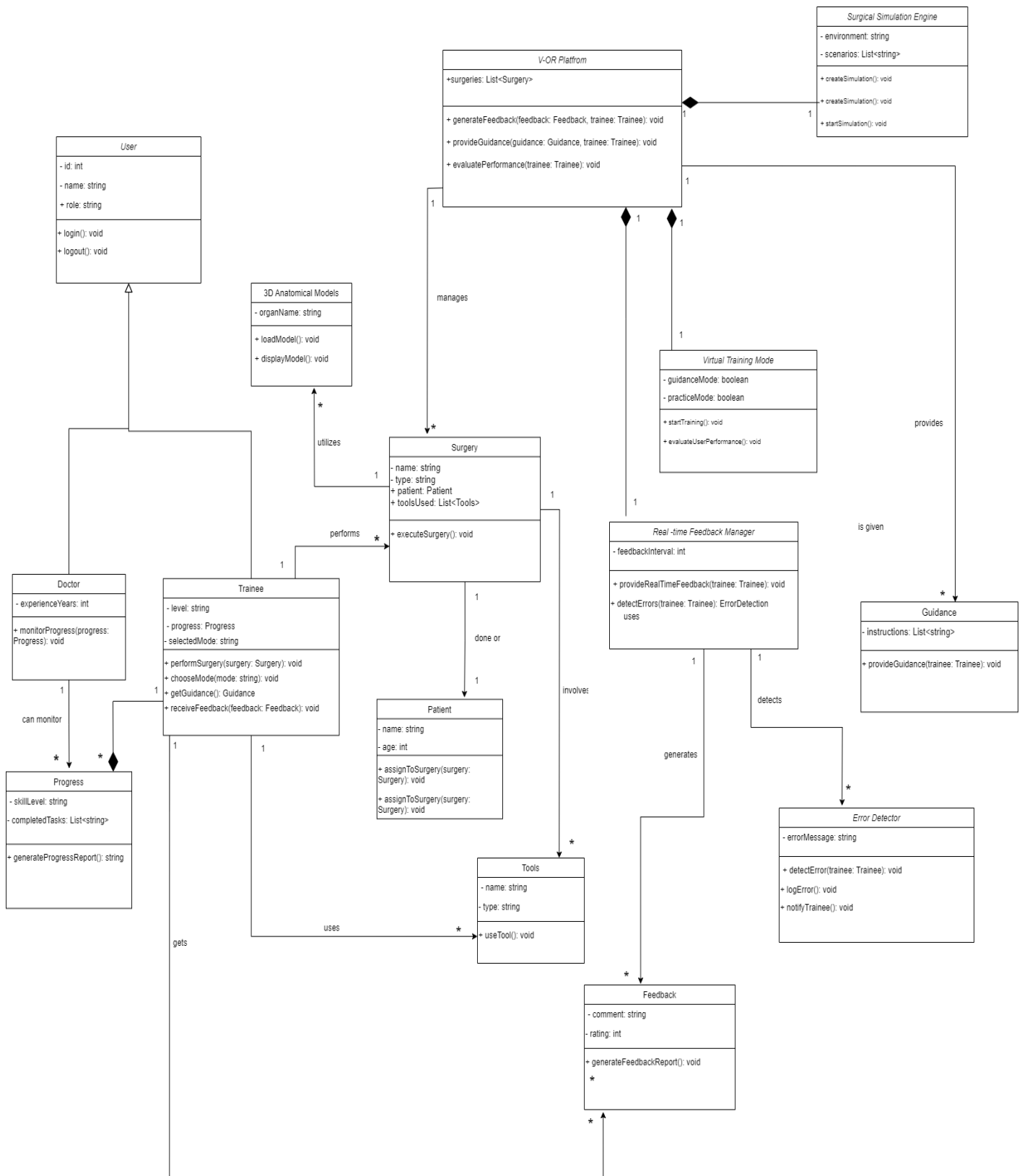


Figure 5: Class Diagram

- **Class level Sequence Diagram**

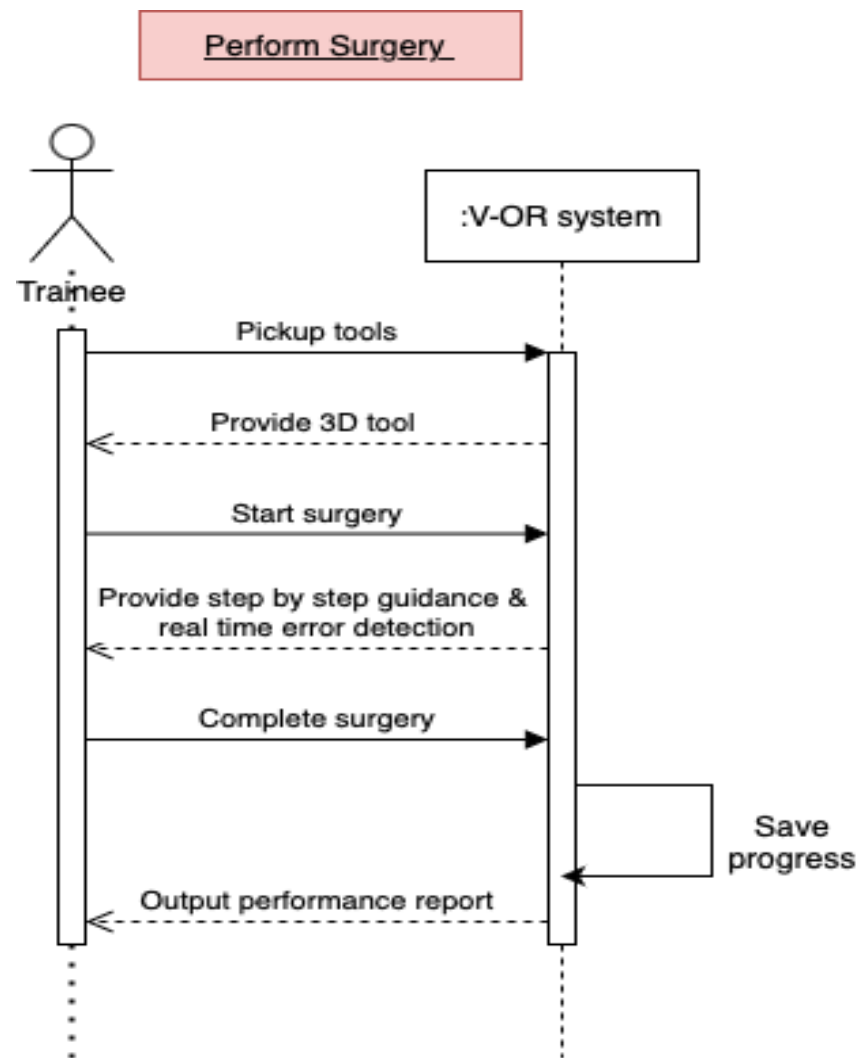


Figure 6: SD Perform Surgery

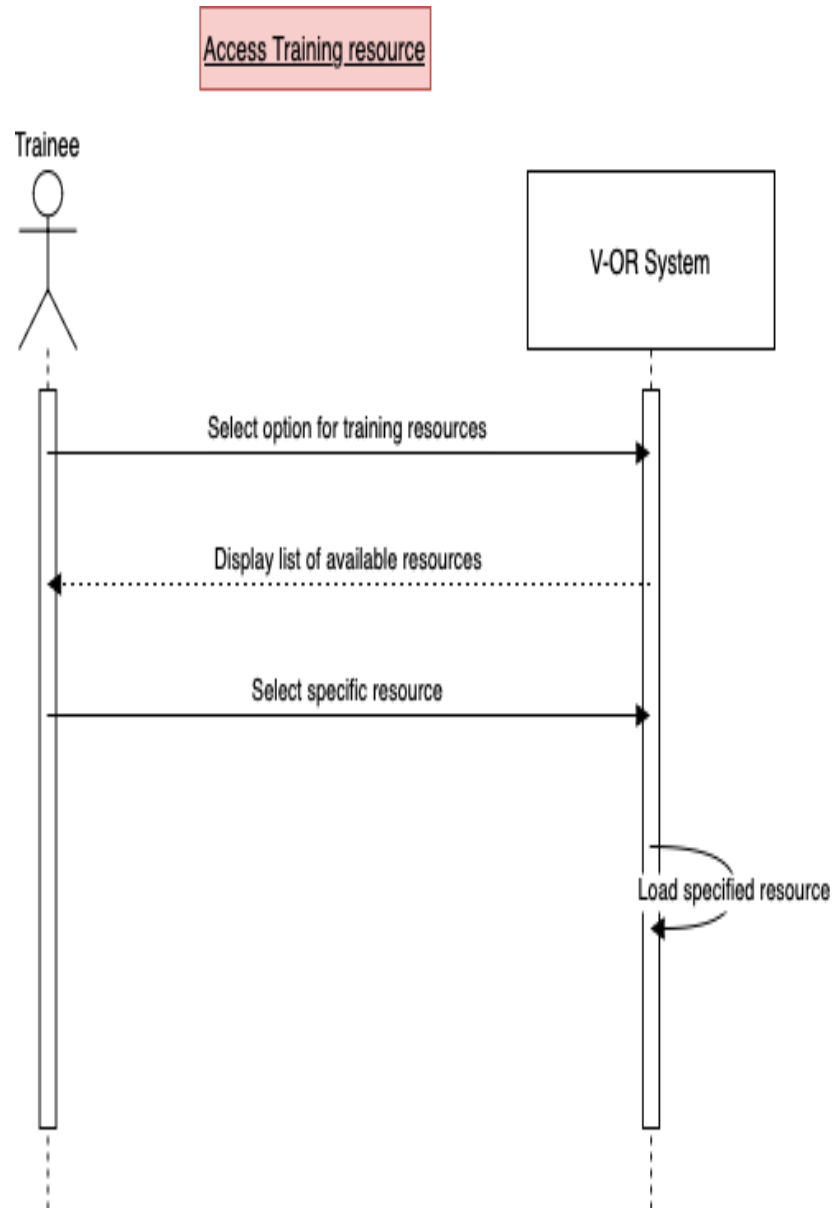


Figure 7: SD Access Training Resources

Receive Guidance

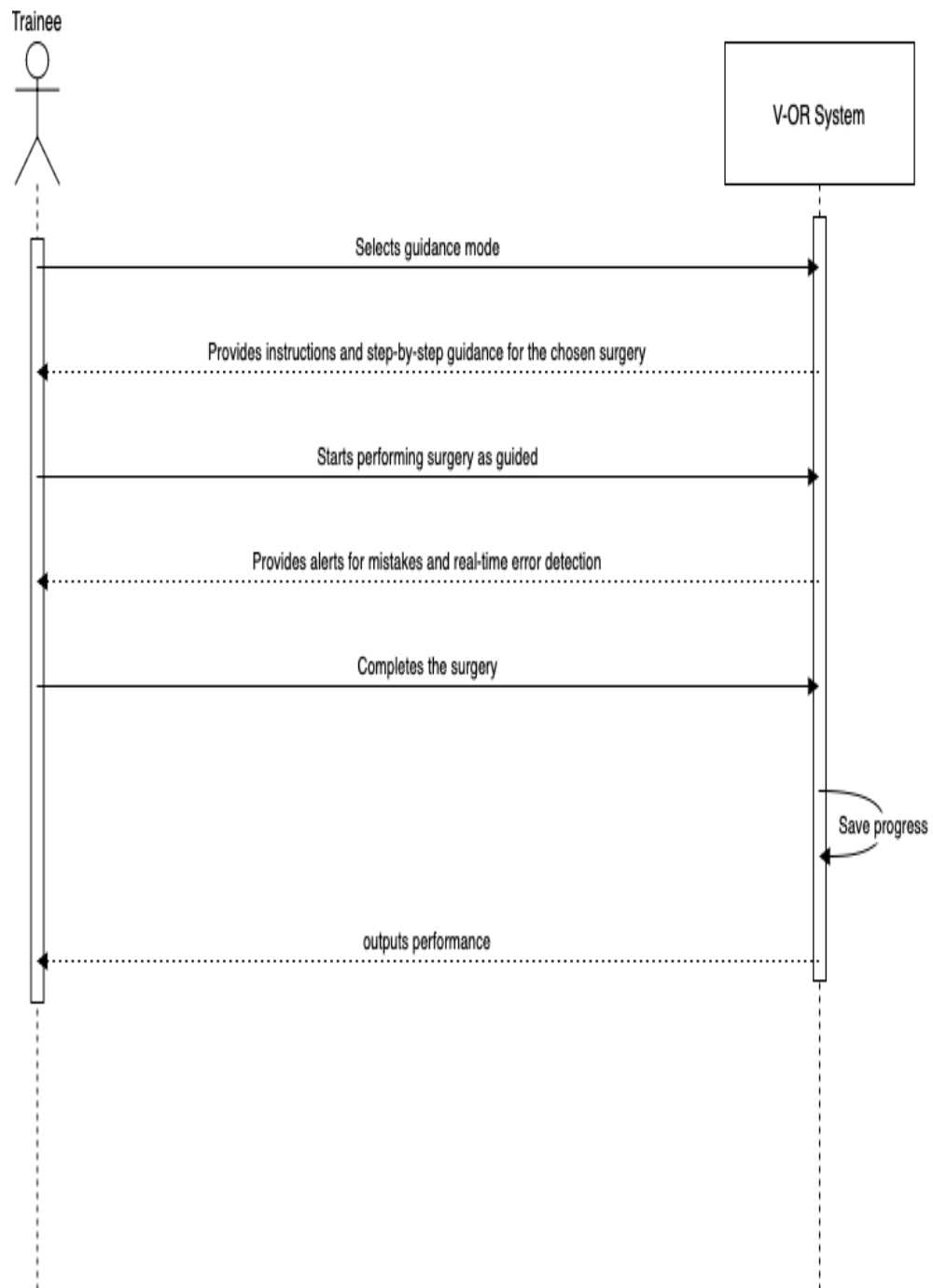


Figure 8: SD Receive Guidance

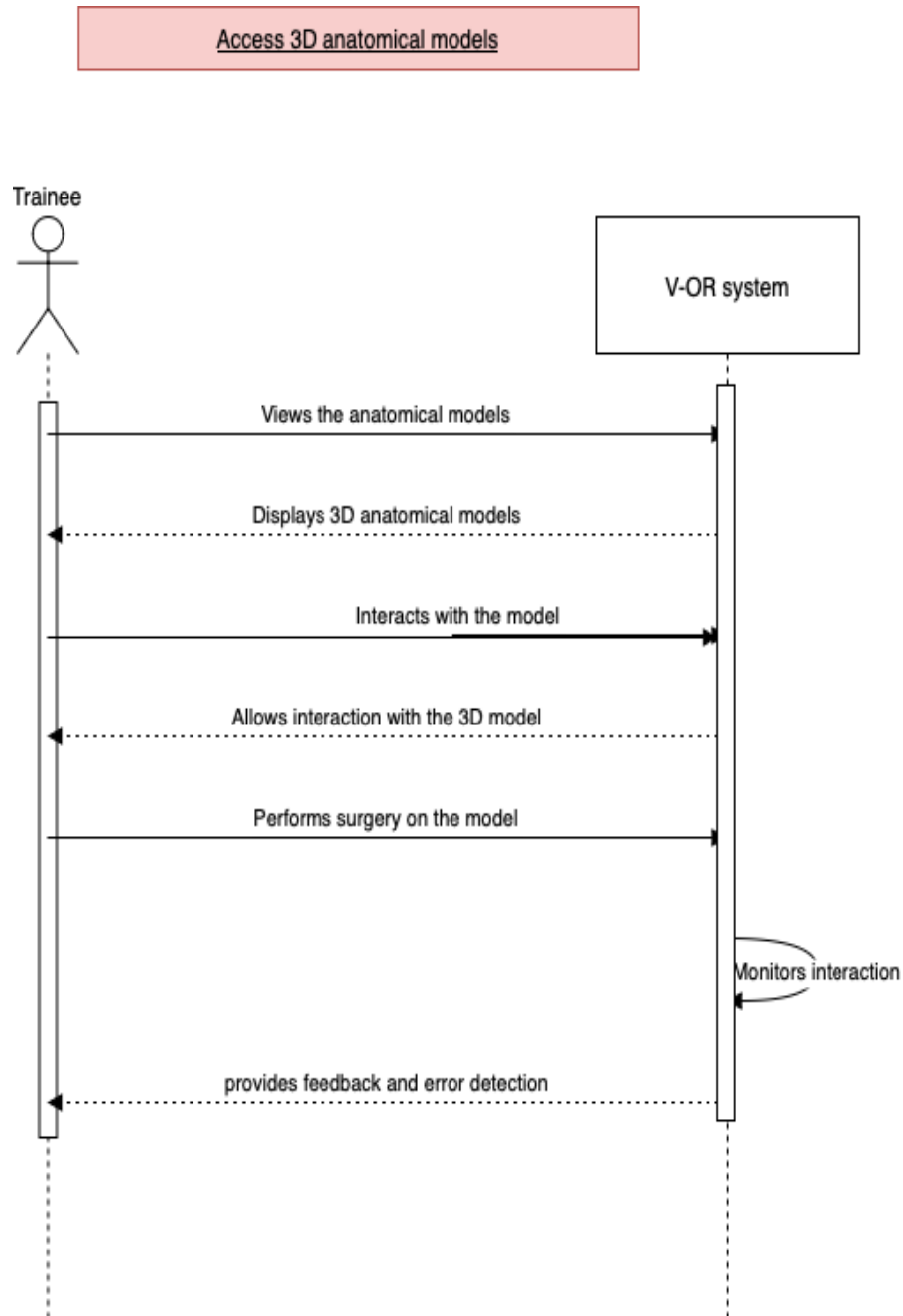


Figure 9: SD Access 3D Models

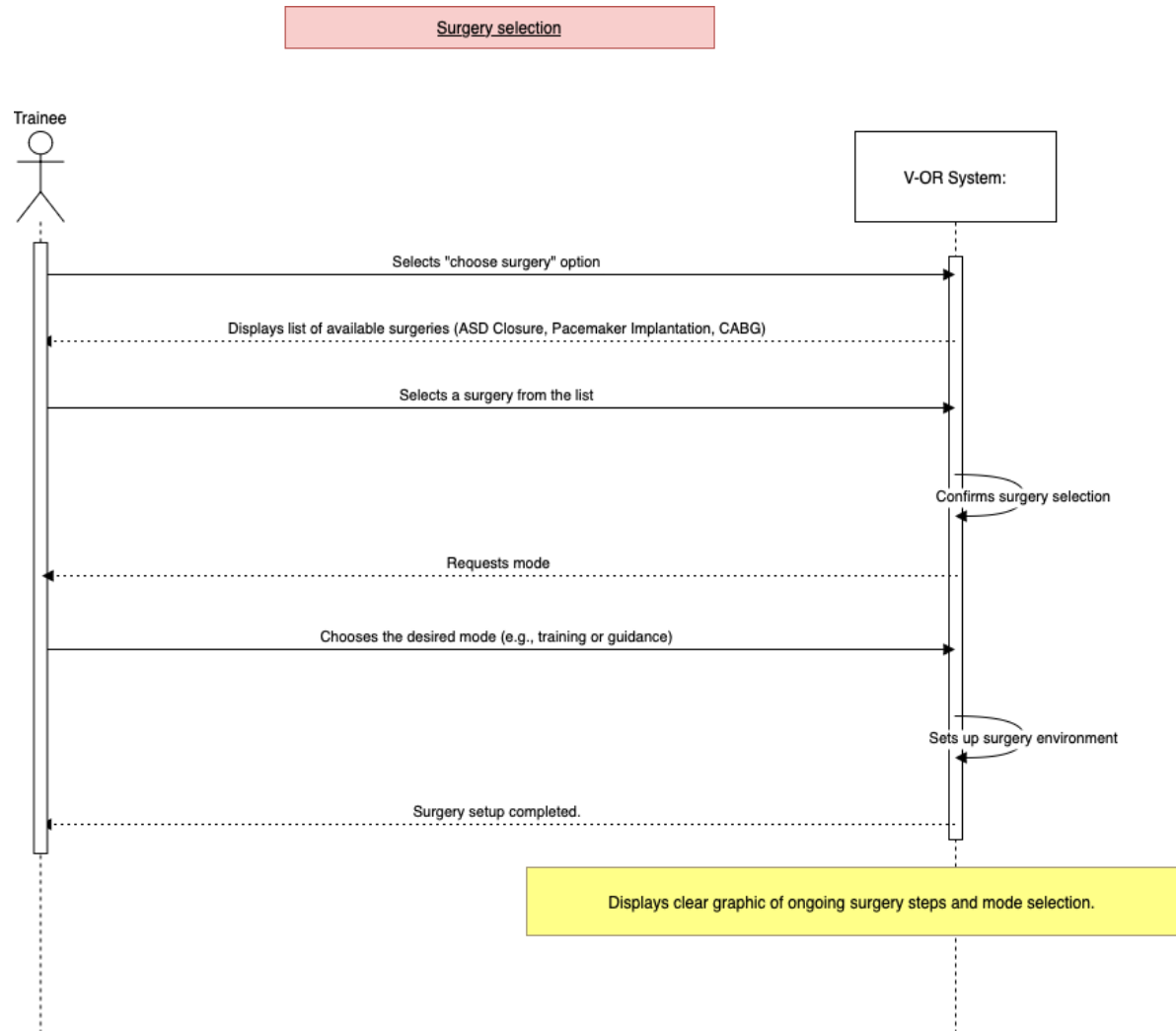
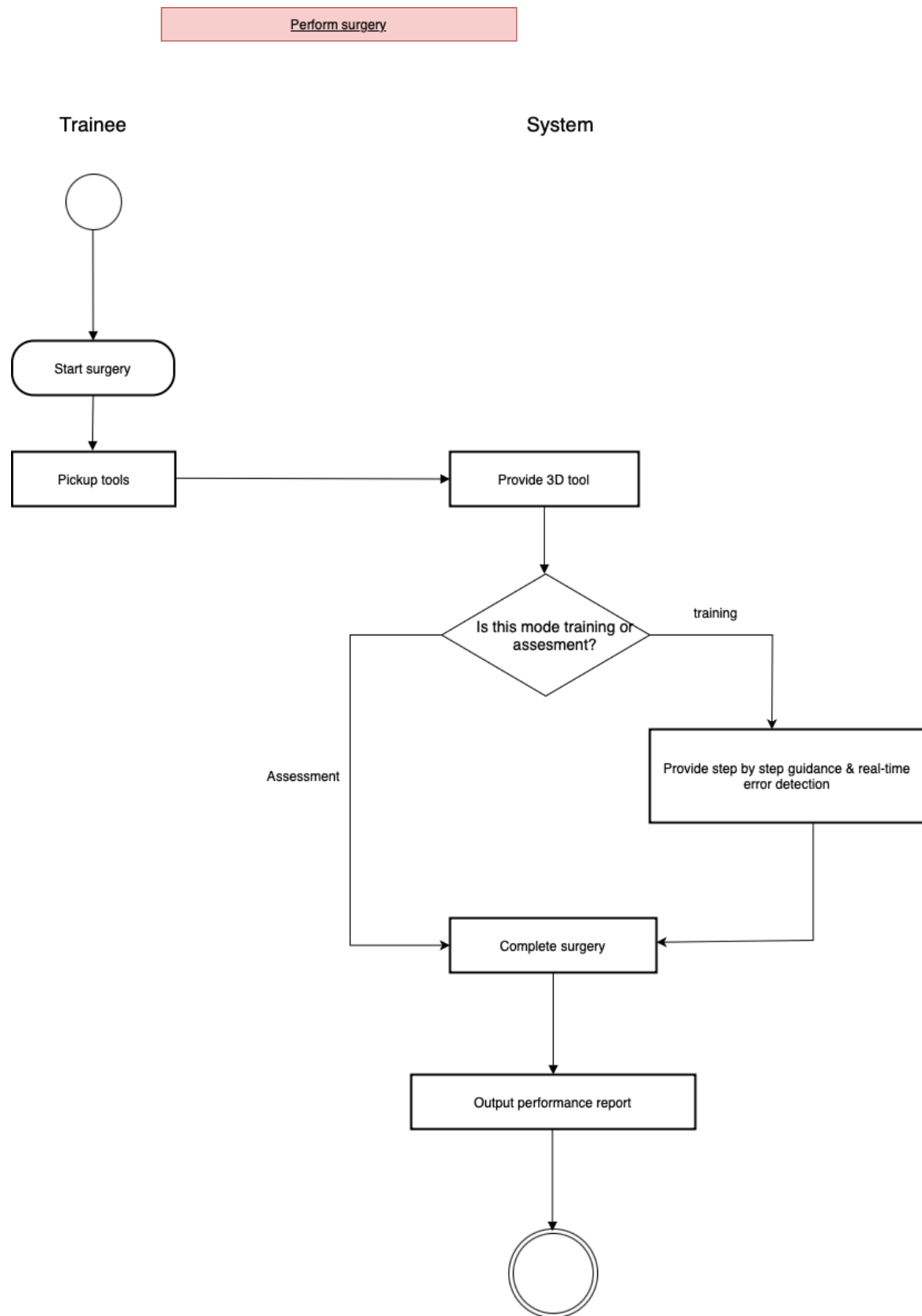
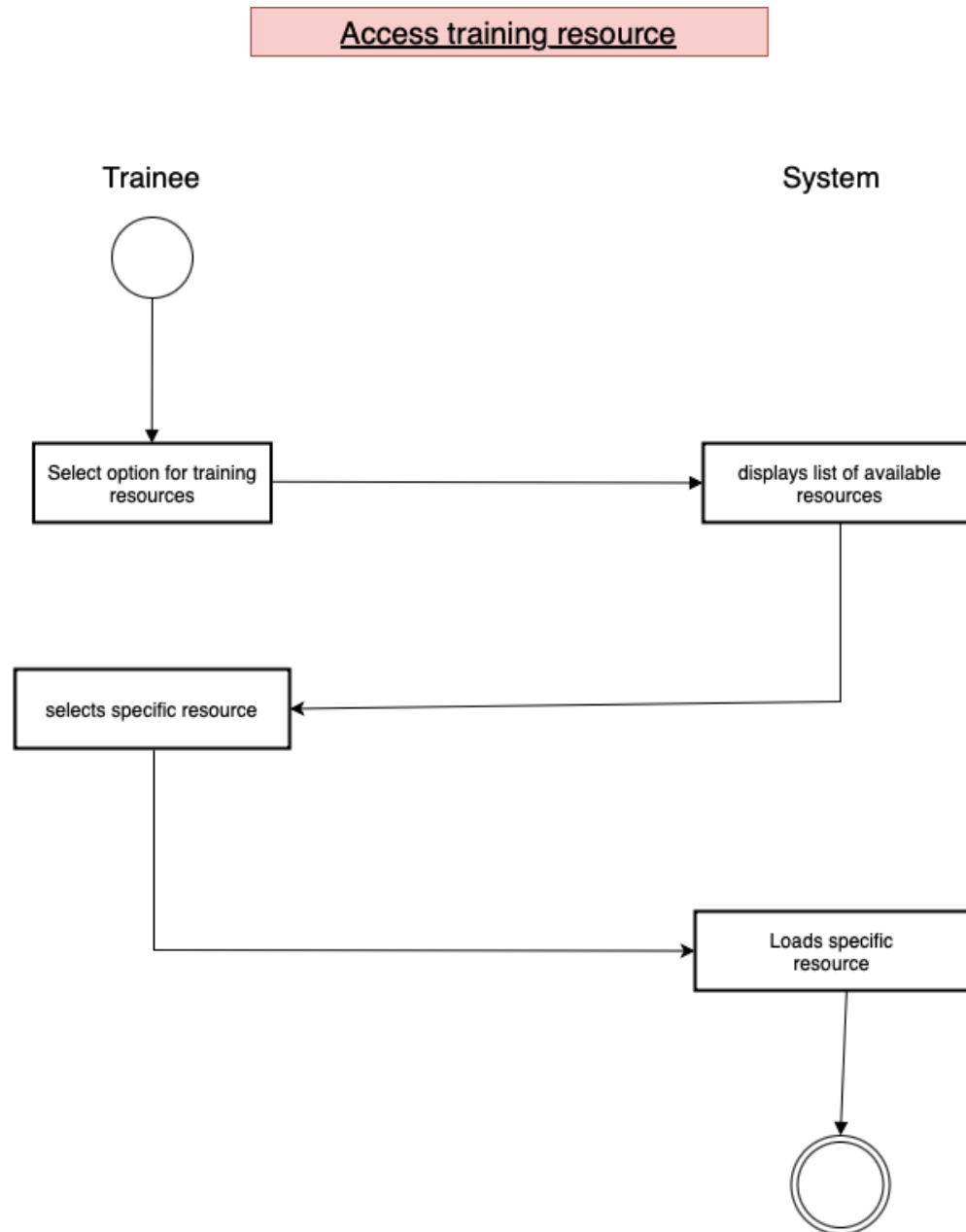


Figure 10: SD Surgery Selection

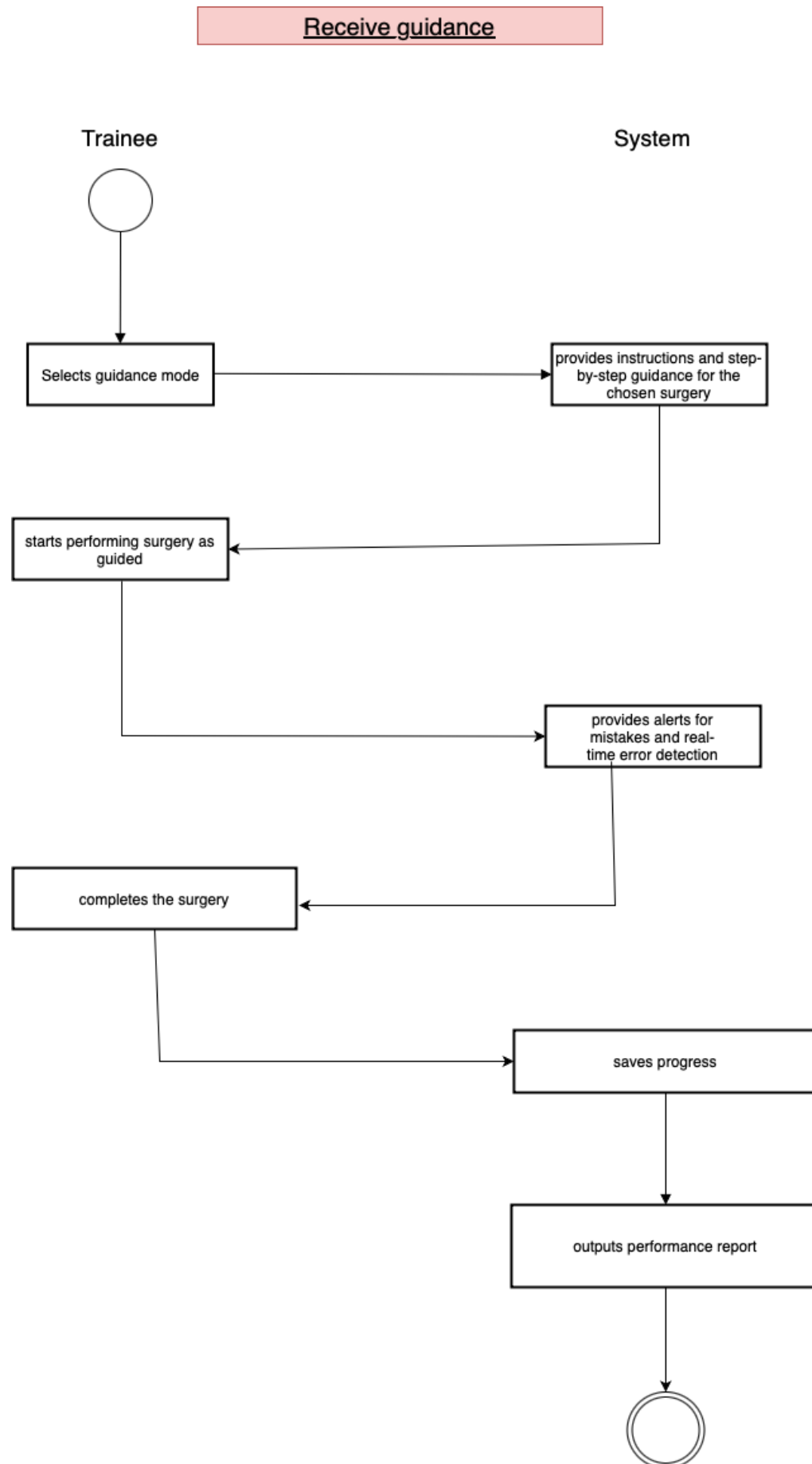
- Activity Diagrams



Activity Diagram 1

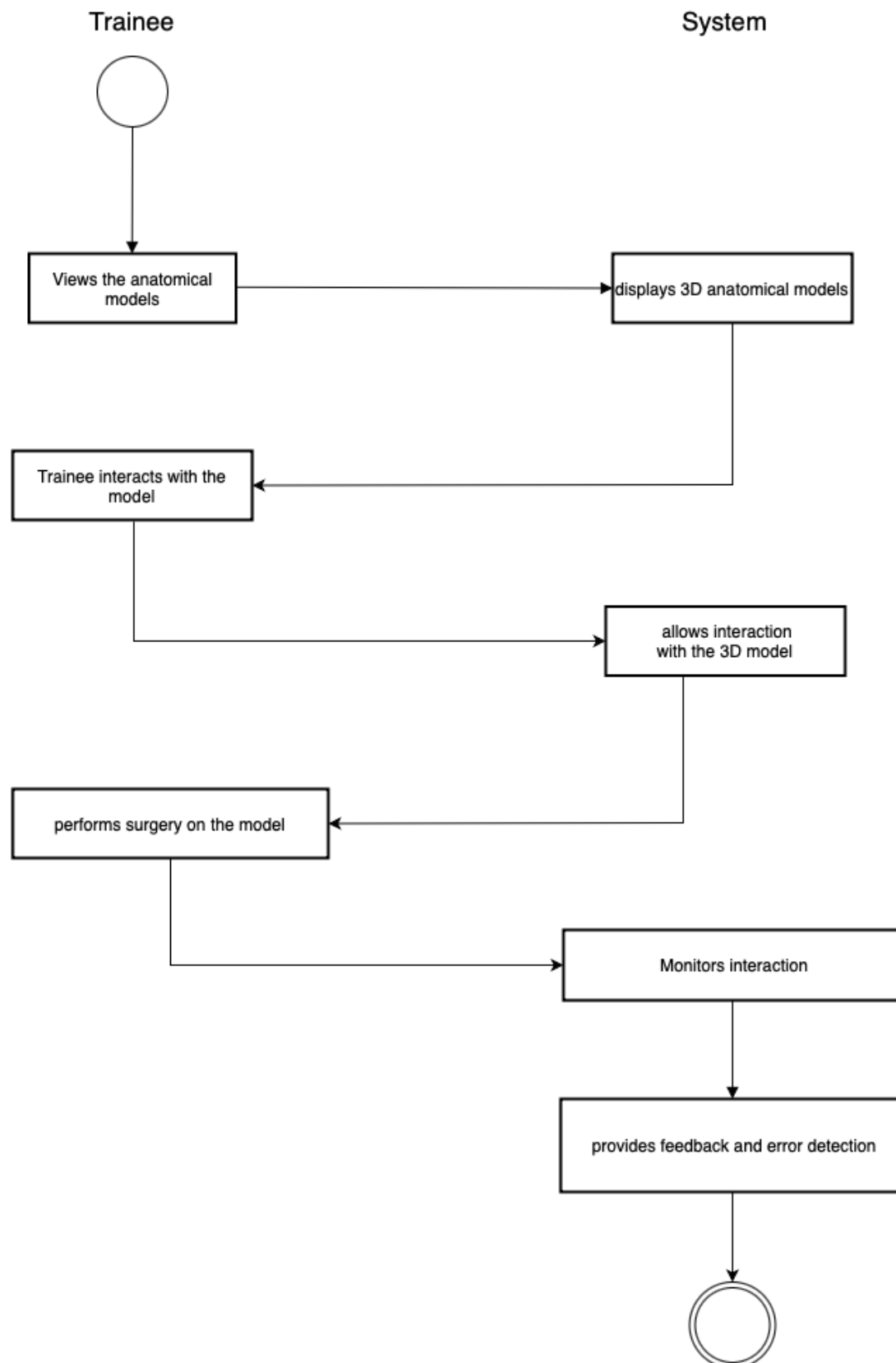


Activity Diagram 2

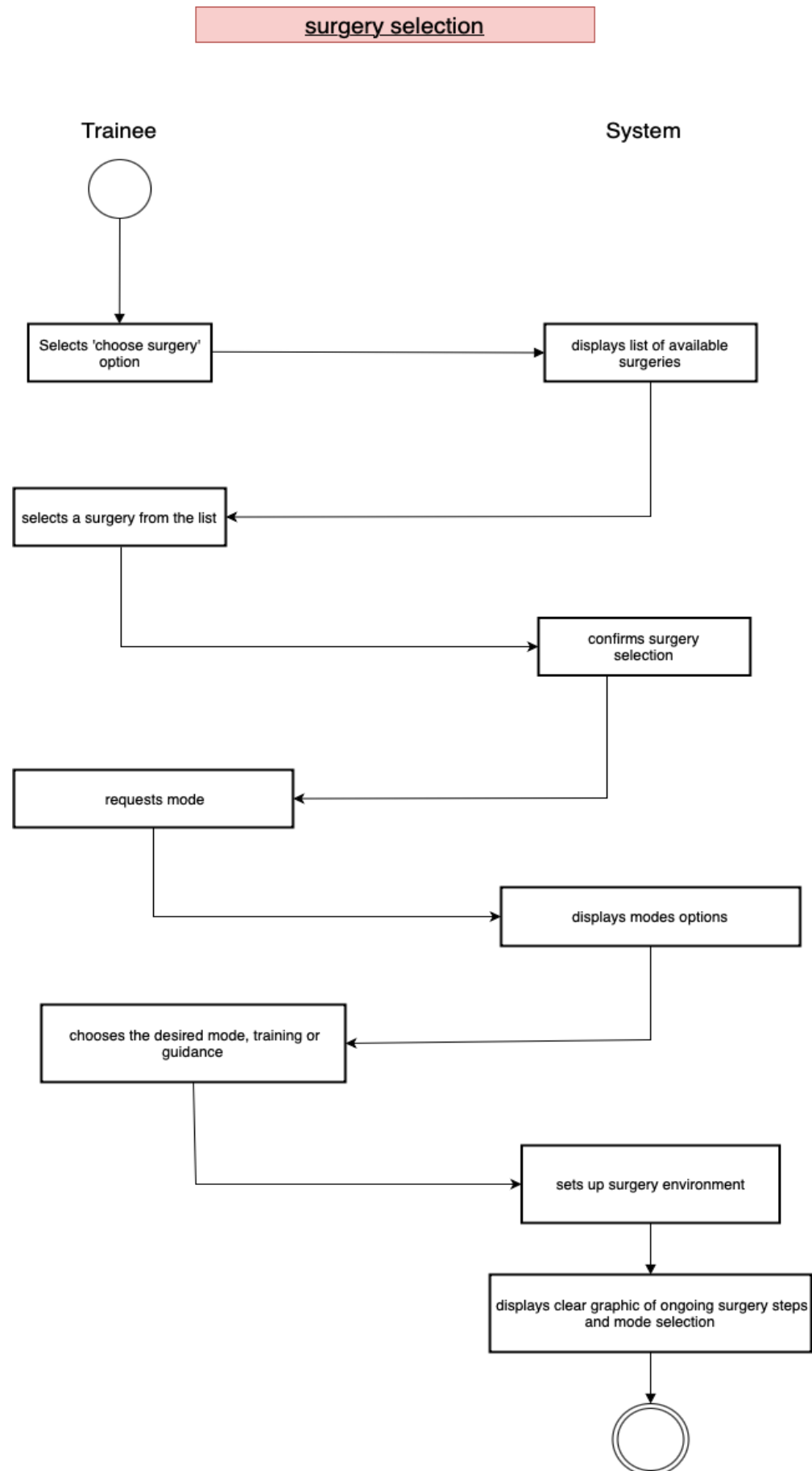


Activity Diagram 3

Access 3D models

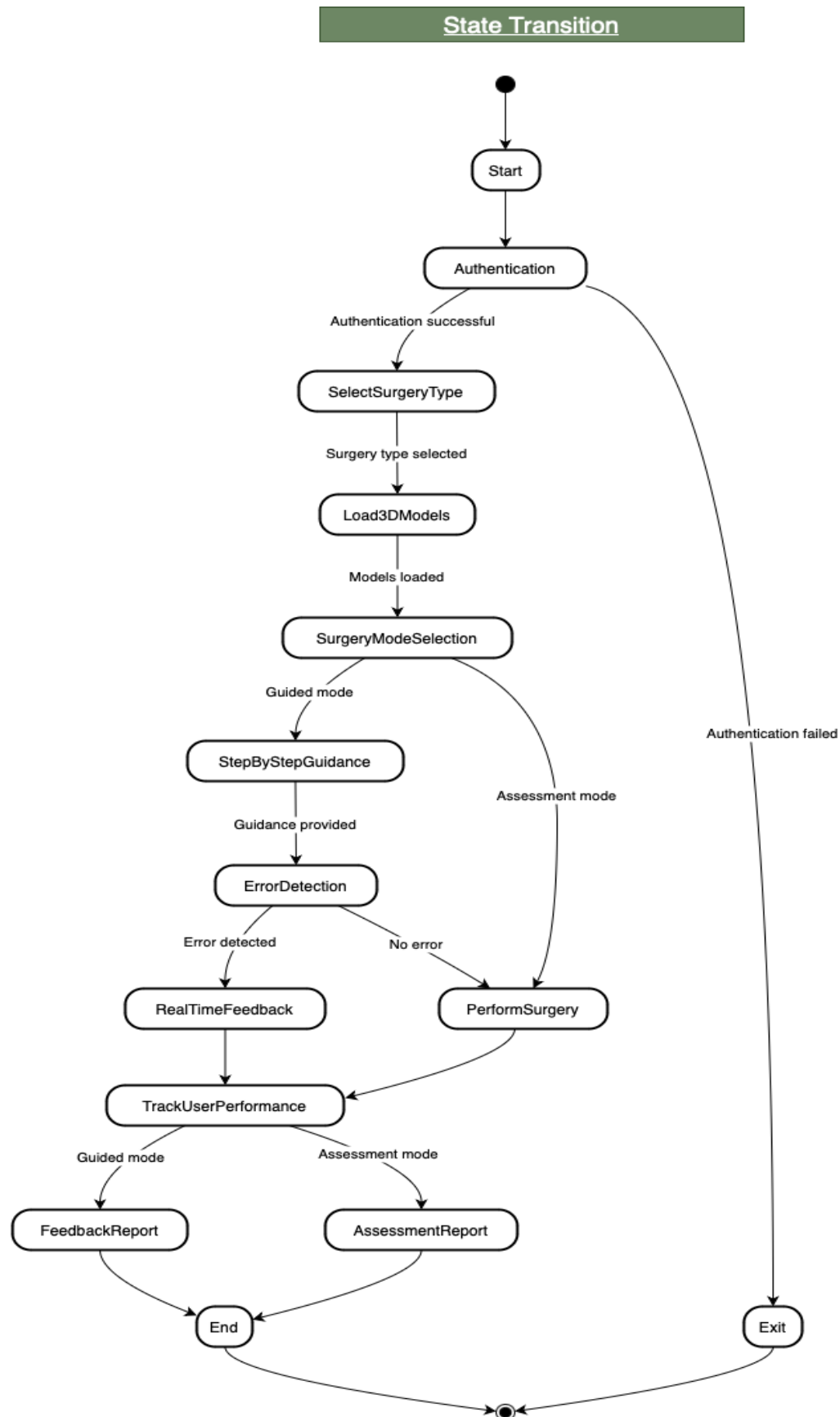


Activity Diagram 4



Activity Diagram 5

- State- Transition Diagram



State-Transition Diagram 1

3.3 Data Design

The information domain of the V-OR system is then converted to a form of structured data entitles stored in a MySQL data base. These tables contain significant entities like User, Surgery, Progress, Feedback ,etc. Like many other resources, the 3D anatomical models used during surgeries in operations plans are file-based and their paths are stored in the database for ease of access to the specific files at the time of simulations. This architecture must maintain data consistency and will perform data processing without inefficiency.

Databases and Data Storage:

MySQL Database: Database preserves tables such as users, surgeries, progress and feedback and etc.

File System on Server: Saves the actual 3D anatomical models.

ERD Diagram:

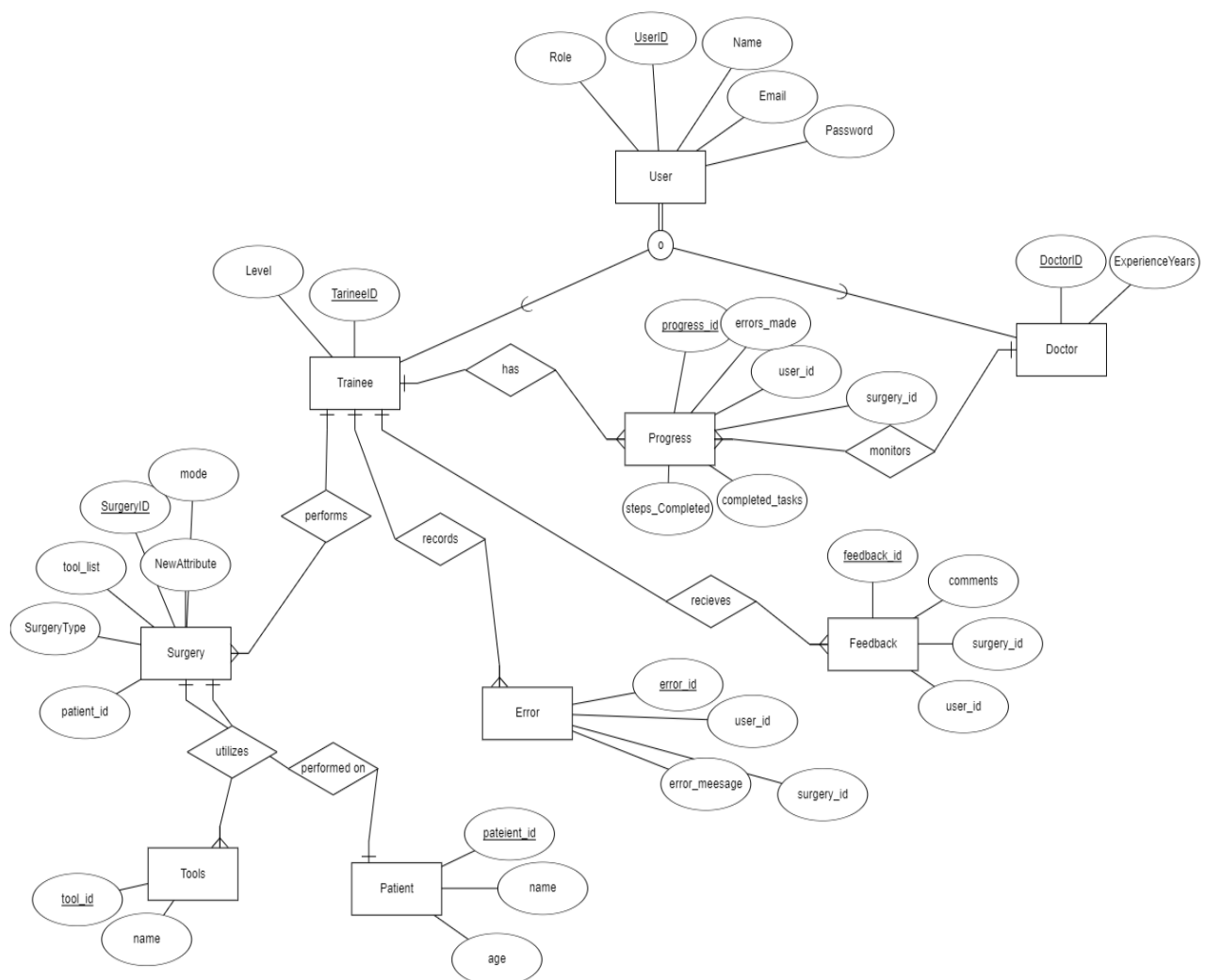


Figure 11: ERD Diagram

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